Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```
SELECT*
FROM `project-big-query-493417.Query.Sales`
LIMIT 1000;


---Q1 (WHERE CLAUSE) Filter all transactions that occurred in the year 2023
SELECT *
FROM `project-big-query-493417.Query.Sales`
WHERE EXTRACT(YEAR FROM Date) = 2023;
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Date	Customer ID	Gender	Age	Produ
1	191	2023-10-18	CUST191	Male	64	Beaut
2	204	2023-09-28	CUST204	Male	39	Beaut
3	230	2023-04-23	CUST230	Male	54	Beaut
4	232	2023-02-06	CUST232	Female	43	Beaut

Results per page: 50 | 1 – 50 of 998

Sales | *Untitled...ery

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```
---Display all transactions where the Total Amount is more than the average Total Amount of the entire dataset.
---Expected Output: All columns

SELECT *
FROM `project-big-query-493417.Query.Sales`
WHERE `Total Amount` > (
  SELECT AVG(`Total Amount`)
  FROM `project-big-query-493417.Query.Sales`);
```

Query completed

Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Age	Product Category	Quantity	Price per Unit	Total Amount
50	Beauty	1	500	500
43	Beauty	1	500	500
25	Beauty	1	500	500
64	Beauty	1	500	500

Sales | *Untitled...ery

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```
--Q3 Agrregate functions
--Calculate the total revenue (sum of Total Amount).

SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `project-big-query-493417.Query.Sales`;

----Q4 Distinct
---Display all distinct Product Categories in the dataset
SELECT DISTINCT(`Product Category`)
FROM `project-big-query-493417.Query.Sales`;
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Total_Revenue
1	456000

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```
SELECT SUM(`Total Amount`) AS Total_Revenue
FROM `project-big-query-493417.Query.Sales`;

----Q4 Distinct
---Display all distinct Product Categories in the dataset
SELECT DISTINCT(`Product Category`)
FROM `project-big-query-493417.Query.Sales`;
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category
1	Beauty
2	Clothing
3	Electronics

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```google_sql
---Q5 Group by
---For each Product Category, calculate the total quantity sold.
SELECT
  `Product Category`,
  SUM(Quantity) AS `total_quantity_sold`
FROM `project-big-query-493417.Query.Sales`
GROUP BY `Product Category`
ORDER BY `total_quantity_sold` DESC;

---Q6 CASE Statement
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Product Category	total_quantity_sold
1	Clothing	894
2	Electronics	849
3	Beauty	771

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```google_sql
---Create a column called Age_Group that classifies customers as:
--- 'Youth' (<30) 'Adult' (30-59)
--- 'Senior' (60+)


SELECT `Customer ID`, Age,
CASE
WHEN Age <30 THEN 'Youth'
WHEN Age BETWEEN 30 and 59 THEN 'Adult'
WHEN Age>= 60 THEN'Senior'
END AS `Age Group`
FROM `project-big-query-493417.Query.Sales`;
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Age Group
6	CUST310	28	Youth
7	CUST363	64	Senior

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```
FROM `project-big-query-493417.Query.Sales`;

----Q7 Conditional Aggregation
----For each Gender, count how many high-value transactions occurred (where Total Amount > 500).

SELECT `Gender`, COUNT(`Transaction ID`) AS High_Value_transaction
FROM `project-big-query-493417.Query.Sales`
WHERE `Total Amount` > 500
GROUP BY `Gender`;

----Q8 HAVING CLAUSE
---For each Product Category, show only those categories where the total revenue exceeds 5,000
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Ope

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Gender	High_Value_trans...
1	Female	155
2	Male	144

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```
GROUP BY `Gender`;

----Q8 HAVING CLAUSE
---For each Product Category, show only those categories where the total revenue exceeds 5,000

SELECT `Product Category` AS product_category, SUM(`Total Amount`) AS total_revenue
FROM `project-big-query-493417.Query.Sales`
GROUP BY `Product Category`
HAVING SUM(`Total Amount`) > 5000;

----Q9 Calculated Fields
---Display a new column called Unit_Cost_Category that labels a transaction as:
---- 'Cheap' if Price per Unit < 50
```

Query completed

Using on-demand processing quota | Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	product_category	total_revenue
1	Beauty	143515
2	Clothing	155580
3	Electronics	156905

Results per page: 50 | 1 – 3 of 3

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```sql
---- 'Cheap' if Price per Unit < 50
--- 'Moderate' if Price per Unit between 50 and 200
--- 'Expensive' if Price per Unit > 200


SELECT `Transaction ID`, `Price per unit`,
CASE
  WHEN `Price per Unit`< 50 THEN 'Cheap'
  WHEN `Price per Unit`BETWEEN 50 AND 200 THEN 'Moderate'
  WHEN `Price per Unit`> 200 THEN 'Cheap'
  END AS `Cost Category`
FROM `project-big-query-493417.Query.Sales`;
```

Query completed

Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Transaction ID	Price per unit	Cost Category
35	25	50	Moderate
36	27	25	Cheap
37	40	50	Moderate

Sales | *Untitled...ery

Untitled query | Run | Save | Download | Share | Schedule

```sql
----Display all transactions from customers aged 40 or older and add a column Spending_Level showing:
----- 'High' if Total Amount > 1000
---- 'Low' otherwise


  SELECT `Customer ID`, Age, `Total Amount`,
CASE
  WHEN `Total Amount` >1000 THEN 'High'
  ELSE 'Low'
  END AS Spending_Level
FROM `project-big-query-493417.Query.Sales`
  WHERE Age >=40
```

Query completed

Processing location: africa-south1

Query results | + Create conversation | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Execution graph

Row	Customer ID	Age	Total Amount	Spending_Level
20	CUST744	40	25	Low
21	CUST047	40	1500	High